

Hints & Solutions

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Q1 (4) - Single Correct

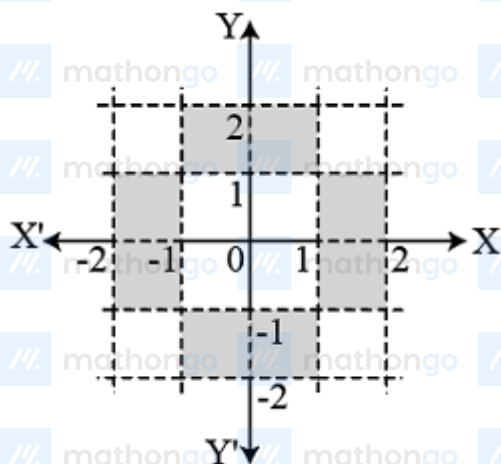
If $[|x|] = 1$ and $[|y|] = 0$

Then, $1 \leq |x| < 2, 0 \leq |y| < 1$

$\Rightarrow x \in (-2, -1] \cup [1, 2), y \in (-1, 1)$

If $[|x|] = 0, [y] = 1$

Then, $x \in (-1, 1)$ and $y \in (-2, -1] \cup (1, 2]$

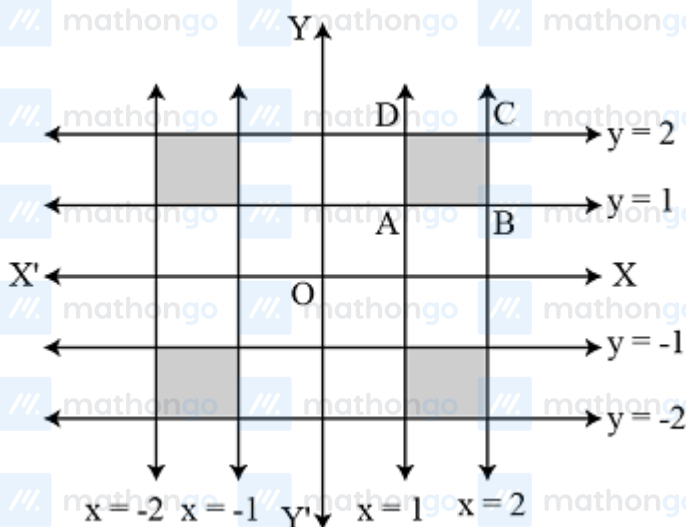


Hence, required area $= 4(2 - 1)[1 - (-1)] = 8$ units

Q2 (1) - Single Correct

$\min\{|x|, |y|\} = 1$

$\max\{|x|, |y|\} = 2$



$\therefore$ Required area $= 4 \times \text{Area of rectangle } (ABCD)$
 $= 4 \times (1) = 4\text{sq units}$

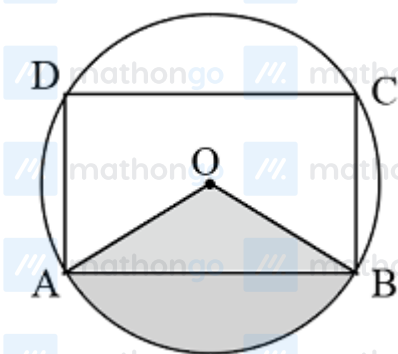
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Q3 (1) - Single Correct

The shaded area is the required region



$$= \frac{\pi r^2}{4} = 4\pi \text{ sq units}$$

Q4 (2) - Single Correct

Given, $ay^2 = x^2(a - x) \Rightarrow y = \pm x\sqrt{\frac{a-x}{a}}$

For the tracing of the curve, we must have $x \leq a$.

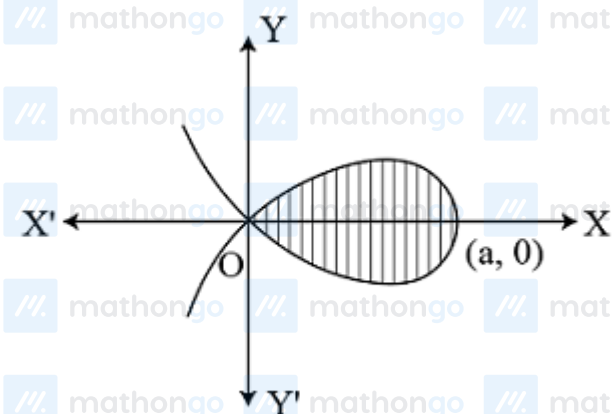
For $0 < x \leq a, y > 0$ and for $x < 0, y < 0$

Also, $y = 0 \Rightarrow x = 0, a$

Curve is symmetrical about X-axis,

when $x \rightarrow -\infty, y \rightarrow -\infty$

Also, it can be verified that y has only one point of maxima for $0 < x < a$



Required area $= 2 \int_0^a x\sqrt{\frac{a-x}{a}} dx$

Put $\sqrt{\frac{a-x}{a}} = t \Rightarrow 1 - \frac{x}{a} = t^2$

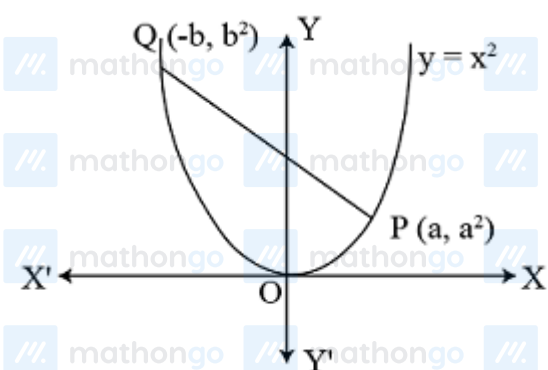
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$$\begin{aligned} \Rightarrow x &= a(1-t^2) \\ \Rightarrow dx &= -2at dt \\ \therefore A &= 2 \int_1^0 a(1-t^2)(t)(-2at) dt \\ &= 4a^2 \int_0^1 (t^2 - t^4) dt \\ &= 4a^2 \left[\frac{t^3}{3} - \frac{t^5}{5} \right]_0^1 \\ &= 4a^2 \left[\frac{1}{3} - \frac{1}{5} \right] = \frac{8a^2}{15} \text{ sq. unit} \end{aligned}$$

Q5 (1) - Single Correct

Given, $y = x^2$... (i)



$$\text{Slope of } PQ = \frac{a^2 - b^2}{a + b} = a - b$$

$$\text{Equation of } PQ \text{ is } y - a^2 = \frac{a^2 - b^2}{a + b}(x - a)$$

$$\Rightarrow y - a^2 = (a - b)(x - a)$$

$$\Rightarrow y = a^2 + x(a - b) - a^2 + ab$$

$$\Rightarrow y = (a - b)x + ab$$

$$\therefore S_1 = \int_{-b}^a [(a - b)x + ab - x^2] dx$$

$$= \left[(a - b) \frac{x^2}{2} + abx - \frac{x^3}{3} \right]_{-b}^a$$

$$= \frac{(a+b)^3}{6} \dots (i)$$

$$\text{Also, } S_2 = \frac{1}{2} \begin{vmatrix} a & a^2 & 1 \\ -b & b^2 & 1 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= \frac{1}{2} [ab^2 + a^2b]$$

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$$= \frac{1}{2}ab(a+b) \dots (ii)$$

$$\text{Now, } \frac{S_1}{S_2} = \frac{(a+b)^3}{6} \cdot \frac{2}{ab(a+b)} = \frac{(a+b)^2}{3ab}$$

$$= \frac{1}{3} \left[\frac{a}{b} + \frac{b}{a} + 2 \right]$$

$$\therefore \left[\frac{S_1}{S_2} \right]_{\min} = \frac{4}{3}$$

Q6 (3) - Single Correct

Given, $y = f(x)$ is function such that

$$f''(0) = 0 \dots (i)$$

$$f''(x) = f'(x) \dots (ii)$$

$$\text{and } f'(0) = 1 \dots (iii)$$

From Eq. (ii), we have

$$\frac{f''(x)}{f'(x)} = 1$$

$$\log f'(x) = x + C$$

$$\therefore f'(0) = 1$$

$$\therefore C = 0$$

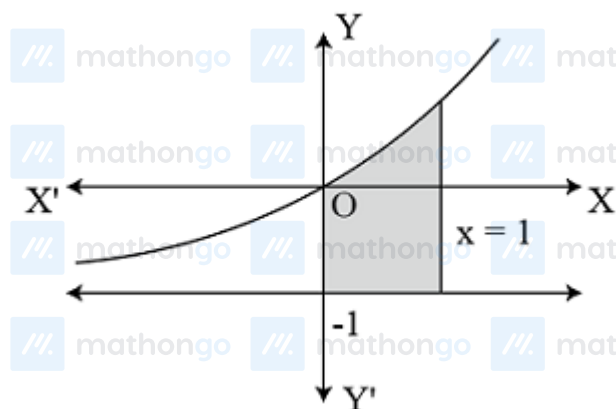
$$\text{Now, } f'(x) = e^x$$

$$\Rightarrow f(x) = e^x + K$$

$$\therefore f(0) = 0$$

$$\therefore K = -1$$

$$\text{Then, } f(x) = e^x - 1$$



$$\therefore \text{Required area} = \int_0^1 (e^x - 1 + 1) dx = [e^x]_0^1 = (e - 1)$$

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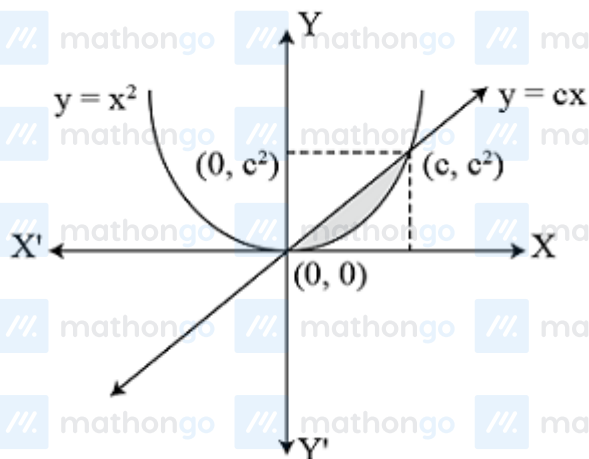
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Q7 (1, 3) - Multiple Correct

$$\text{Area}(T) = \frac{c \cdot c^2}{2} = \frac{c^3}{2}$$

$$\begin{aligned} \text{Area}(R) &= \frac{c^3}{2} - \int_0^c x^2 dx \\ &= \frac{c^3}{2} - \frac{c^3}{3} = \frac{c^3}{6} \end{aligned}$$



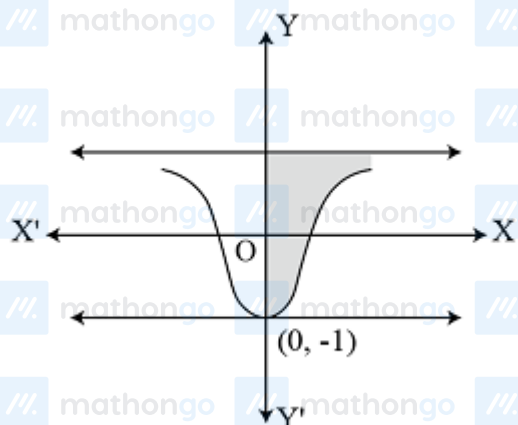
$$\therefore \lim_{c \rightarrow 0^+} \frac{\text{area}(T)}{\text{area}(R)} = \lim_{c \rightarrow 0^+} \frac{\frac{c^3}{2}}{\frac{c^3}{6}} = 3$$

Q8 (1, 3, 4) - Multiple Correct

$$y = f(x) = \frac{x^2 - 1}{x^2 + 1} = 1 - \frac{2}{x^2 + 1}$$

$$f'(x) = \frac{4x}{(x^2 + 1)^2}, x > 0$$

$f(x)$ is increasing and $x < 0$, $f(x)$ is decreasing.



Range is $[-1, 1] \Rightarrow$ into

Minimum value occurs at $x = 0$ and $f(0) = -1$

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$$\therefore A = 2 \int_0^{\infty} \left(1 - \frac{x^2+1}{x^2+1}\right) dx = 4 \int_0^{\infty} \frac{dx}{x^2+1}$$

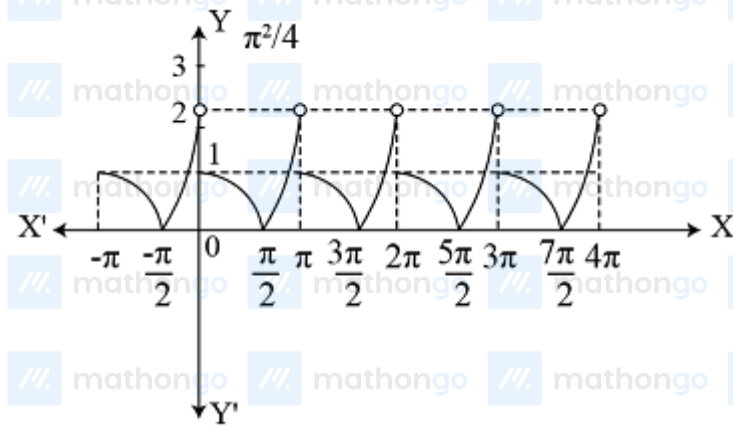
$$= 4 [\tan^{-1} x]_0^{\infty} = 4 \times \frac{\pi}{2} = 2\pi$$

Q9 (1, 4) - Multiple Correct

$$\text{Given, } f(x) = \begin{cases} \cos x, & 0 \leq x < \frac{\pi}{2} \\ \left(\frac{\pi}{2} - x\right)^2, & \frac{\pi}{2} \leq x < \pi \end{cases}$$

and f is periodic with period π

The graph of $f(x)$ could be plotted as



From the graph, the range of function is $\left[0, \frac{\pi^2}{4}\right]$

It is discontinuous at $x = n\pi, n \in I$ It is not differentiable at $x = \frac{n\pi}{2}, n \in I$

Area bounded by $y = f(x)$ and the X -axis from $-n\pi$ to $n\pi$ for $n \in N$.

$$\therefore \text{Area} = 2n \int_0^{\pi} f(x) dx = 2n \left[\int_0^{\pi/2} \cos x dx + \int_{\pi/2}^{\pi} \left(\frac{\pi}{2} - x\right)^2 dx \right]$$

$$= 2n \left(1 + \frac{\pi^3}{24}\right)$$

Q10 (2, 3, 4) - Multiple Correct

$$\text{Given, } f(x) = 2x - x^2 \dots \text{(i)}$$

$$\text{and } g(x) = x^n \dots \text{(ii)}$$

On solving Eqs. (i) and (ii), we get

$$2x - x^2 = x^n$$

$$\Rightarrow x = 0 \text{ and } x = 1$$

$$\therefore \text{Area (A)} = \int_0^1 (2x - x^2 - x^n) dx$$

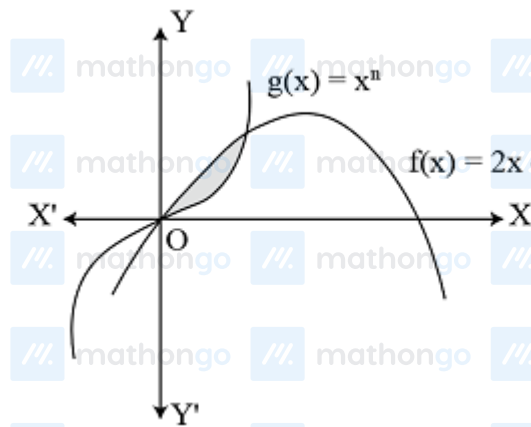
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$$= \left[x^2 - \frac{x^3}{3} - \frac{x^{n+1}}{n+1} \right]_0^1$$

$$= 1 - \frac{1}{3} - \frac{1}{n+1} = \frac{2}{3} - \frac{1}{n+1}$$



According to the question,

$$\frac{2}{3} - \frac{1}{n+1} = \frac{1}{2}$$

$$\Rightarrow \frac{2}{3} - \frac{1}{2} = \frac{1}{n+1}$$

$$\Rightarrow \frac{4-3}{6} = \frac{1}{n+1}$$

$$\Rightarrow n+1 = 6$$

$$\Rightarrow n = 5$$

Thus n is divisor of 15,20 and 30.

Q11 (1, 3) - Multiple Correct

Given, $|x - a_i| + |y| = b_i$, where $i \in N$

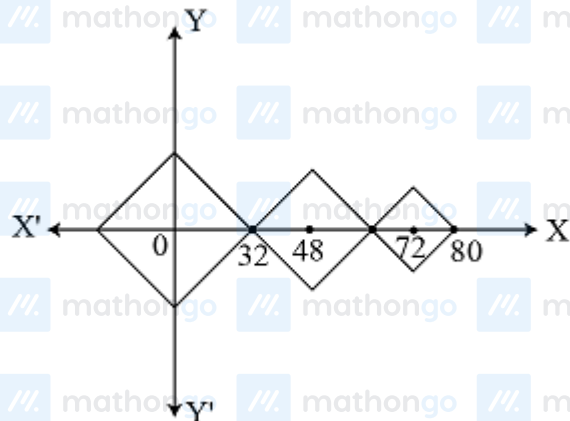
$$\text{where } a_{i+1} = a_i + \frac{3}{2}b_i$$

$$\text{and } b_{i+1} = \frac{b_i}{2}$$

$$\Rightarrow a_1 = 0, b_1 = 32$$

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Now $a_2 = a_1 + \frac{3}{2}b_1 = 4B_1$

and $b_2 = \frac{b_1}{2} = 16$

$\Rightarrow a_3 = 48 + \frac{3}{2} \times 16 = 72$

and $b_3 = \frac{16}{2} = 8$

So, the three loops from $i = 1$ to $i = 3$ are alike

Now, area of the loop (square) $= \frac{1}{2} (\text{Diagonal})^2$

$A_i = \frac{1}{2} (2b_i)^2 = 2(b_i)^2$

So, $\frac{A_{i+1}}{A_i} = \frac{2(b_{i+1})^2}{2(b_i)^2} = \frac{1}{4}$

So, the areas form a geometric progression

$\therefore$ The sum of the GP upto infinite terms is

$\lim_{n \rightarrow \infty} \sum_{i=1}^n A_i = A_1 \cdot \frac{1}{1-r}$

$= 2(32)^2 \times \frac{1}{1-\frac{1}{4}}$

$= 2 \times (32)^2 \times \frac{4}{3}$

$= \frac{8}{3} (32)^2 \text{ sq units}$

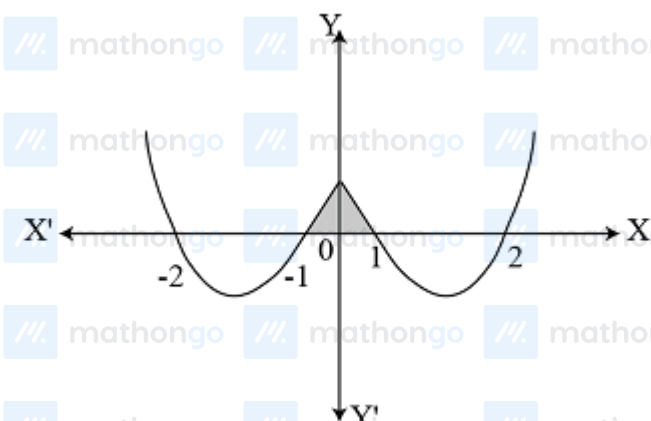
Also, $A_3 = 2(b_3)^2$
 $= 2 \times (8)^2 = 128$

Q12 (1) - Paragraph 1

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The graph of $f(|x|)$ is



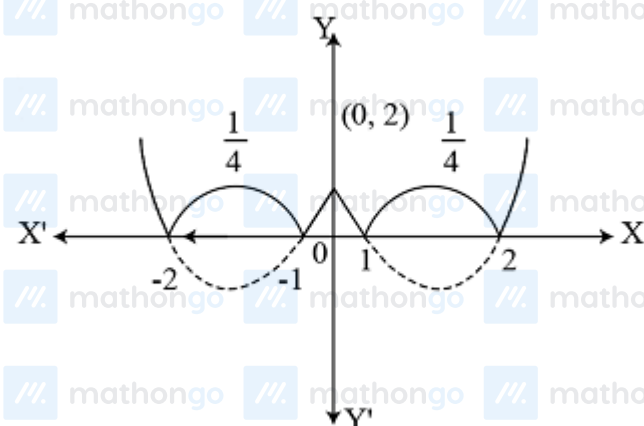
$$\begin{aligned} \therefore \text{Required area} &= 2 \int_0^1 f(x) dx + 2 \left| \int_1^2 f(x) dx \right| \\ &= 2 \int_0^1 (x^2 - 3x + 2) dx + 2 \left| \int_1^2 (x^2 - 3x + 2) dx \right| \\ &= 2 \left(\frac{1}{3} - \frac{3}{2} + 2 \right) + \frac{1}{3} \\ &= \frac{5}{3} + \frac{1}{3} \\ &= 2 \text{ sq units} \end{aligned}$$

Q13 (3) - Paragraph 1

$$\because 1 \leq |x| \leq 2$$

$$\therefore x \in (-2, -1] \cup [1, 2]$$

The graph of $|f(|x|)|$ is



$$\text{Hence, required area} = 2 \left| \int_1^2 f(x) dx \right|$$

$$\begin{aligned} &= 2 \left| \int_1^2 (x^2 - 3x + 2) dx \right| \\ &= 2 \left| \left\{ \frac{x^3}{3} - \frac{3x^2}{2} + 2x \right\}_1^2 \right| \end{aligned}$$

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$$= 2 \left| \left\{ \left(\frac{8}{3} - \frac{12}{2} + 4 \right) - \left(\frac{1}{3} - \frac{3}{2} + 2 \right) \right\} \right|$$

$$= 2 \left| \frac{2}{3} - \frac{5}{6} \right| = \frac{1}{3} \text{ sq unit}$$

Q14 (5) - Integer Type

$$A = \int_0^1 x^6 (1-x)^7 dx = \int_0^1 x^{7-1} (1-x)^{8-1} dx$$

$$= B(7, 8) \quad \left[\because B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx \right]$$

$$= \frac{\Gamma(7)\Gamma(8)}{\Gamma(7+8)} \quad \left[\because B(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)} \right]$$

$$\therefore A = \frac{6!7!}{14!} \quad \left[\because \Gamma(n+1) = n! \right]$$

$$\Rightarrow A^{-1} = \frac{14!}{6!7!} = 2^3 \cdot 3 \cdot 7 \cdot 11 \cdot 13$$

$\therefore$ The number of prime factors is 5.

Q15 (4) - Integer Type

Given curves are

$$x^2 = 4y \dots (i)$$

$$\text{and } y = \frac{8}{x^2+4} \dots (ii)$$

For point of intersection on solving Eqs. (i) and (ii), we get

$$y = \frac{8}{4y+4} \Rightarrow y = \frac{2}{y+1}$$

$$\Rightarrow y^2 + y = 2$$

$$\Rightarrow y^2 + y - 2 = 0$$

$$\Rightarrow y^2 + 2y - y - 2 = 0$$

$$\Rightarrow y(y+2) - 1(y+2) = 0$$

$$\Rightarrow (y+2)(y-1) = 0$$

$$\Rightarrow y = 1, -2$$

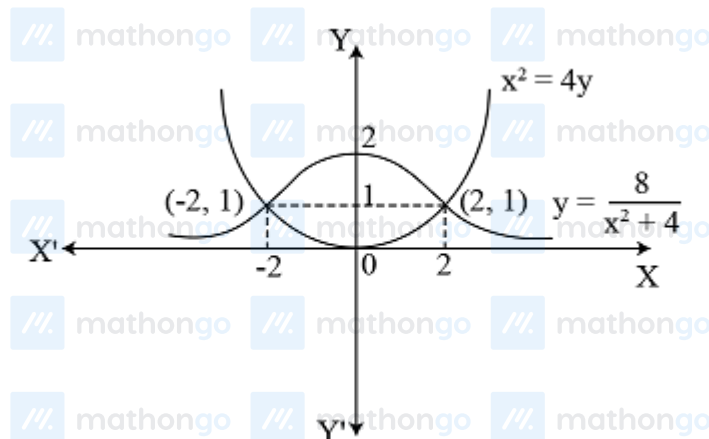
But $y = -2$ is not possible.

When $y = 1$, then $x = \pm 2$

$\therefore$ Points of intersection are $(2, 1)$ and $(-2, 1)$

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$$\therefore \text{Required area} = \int_{-2}^2 \left(\frac{8}{x^2 + 4} - \frac{x^2}{4} \right) dx$$

$$= 8 \left[\tan^{-1} \frac{x}{2} \right]_{-2}^2 - \frac{1}{12} [x^3]_{-2}^2$$

$$= 8 \left[\tan^{-1} 1 - \tan^{-1}(-1) \right] - \frac{1}{12} (8 + 8)$$

$$= 8 \left(\frac{\pi}{4} + \frac{\pi}{4} \right) - \frac{16}{12} = \frac{8\pi}{2} - \frac{16}{12}$$

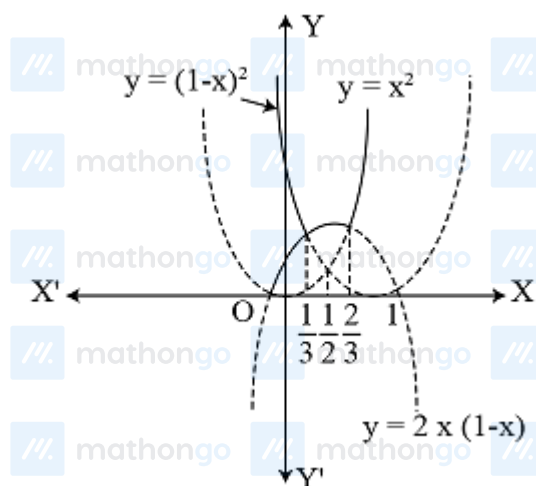
$$= 4\pi - \frac{16}{12} = 4\pi - \frac{4}{3}$$

Hence, $k = 4$

Q16 (7) - Integer Type

We have, $f(x) = \max \{x^2, (1-x)^2; 2x(1-x)\}$

Graphically it could be shown as



From the figure, it is clear that maximum graph i.e. maximum of the given curves are considered and other are neglected.

$$\therefore \text{For } x \in \left[0, \frac{1}{3} \right], x^2 < 2 \times (1-x) \leq (1-x)^2$$

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For $x \in \left[\frac{1}{3}, \frac{1}{2}\right]$, $x^2 \leq (1-x)^2 \leq 2 \times (1-x)$

For $x \in \left[\frac{1}{2}, \frac{2}{3}\right]$, $(1-x)^2 \leq x^2 \leq 2x(1-x)$

For $x \in \left[\frac{2}{3}, 1\right]$, $(1-x)^2 \leq 2x(1-x) \leq x^2$

Hence, $f(x)$ can be written as

$$f(x) = \begin{cases} (1-x)^2, & \text{for } 0 \leq x \leq \frac{1}{3} \\ 2x(1-x), & \text{for } \frac{1}{3} \leq x \leq \frac{2}{3} \\ x^2, & \text{for } \frac{2}{3} \leq x \leq 1 \end{cases}$$

Hence, the area bounded by the curve $y = f(x)$; X -axis and the line $x = 0, x = 1$ is given by

$$\begin{aligned} &= \int_0^{1/3} (1-x)^2 dx + \int_{1/3}^{2/3} 2x(1-x) dx + \int_{2/3}^1 x^2 dx \\ &= \int_0^{1/3} (1+x^2-2x) dx + \int_{1/3}^{2/3} (2x-2x^2) dx + \int_{2/3}^1 x^2 dx \\ &= \left[x + \frac{x^3}{3} - x^2 \right]_0^{1/3} + \left[x^2 - \frac{2x^3}{3} \right]_{1/3}^{2/3} + \left[\frac{x^3}{3} \right]_{2/3}^1 \\ &= \left(\frac{1}{3} + \frac{1}{81} - \frac{1}{9} \right) + \left(\frac{4}{9} - \frac{16}{81} - \frac{1}{9} + \frac{2}{81} \right) + \left(\frac{1}{3} - \frac{8}{81} \right) \\ &= \frac{17}{27} \text{ sq unit} \end{aligned}$$

But area is given $\frac{m}{n}$.

$$\Rightarrow \frac{m}{n} = \frac{17}{27}$$

$$\therefore 2m - n = (2 \times 17 - 27) = 7$$